

HALLOCK, MN, USA

WMO: 727478

Lat: 48.753N

Lon: 96.943W

Elev: 819

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04934

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-18.3	-31.0	1.0	-24.5	-26.0	1.3	-18.3	32.3	9.1	29.5	10.1	6.1	150	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	88.4	71.2	84.4	69.3	82.0	67.9	76.0	83.6	73.4	81.1	71.3	78.9	13.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	127.2	79.9	71.7	120.8	78.8	68.9	109.5	76.1	40.1	84.2	37.6	80.4	35.6	78.3	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.3	25.3	22.6	-28.7	94.6	6.1	2.8	-33.1	96.6	-36.7	98.3	-40.2	99.8	-44.6	101.8
			-28.8	79.7	6.0	2.5	-33.1	81.5	-36.6	82.9	-39.9	84.3	-44.3	86.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	5.9	8.7	23.4	41.3	53.8	64.1	68.1	66.5	58.2	43.8	27.3	13.1
	DBStd	24.60	14.69	14.14	13.96	10.30	9.04	6.69	5.62	6.12	8.61	9.28	11.48	13.06
	HDD50	5797	1366	1156	827	293	64	2	0	0	23	240	682	1144
	HDD65	9589	1831	1576	1290	712	361	93	33	55	237	660	1132	1609
	CDD50	2029	0	0	2	31	182	426	562	511	268	46	1	0
	CDD65	346	0	0	0	0	14	67	130	101	32	2	0	0
	CDH74	4033	0	0	0	18	265	753	1353	1195	421	28	0	0
	CDH80	1198	0	0	0	2	78	208	402	382	122	4	0	0
Wind	WSAvg	10.1	10.5	10.4	10.8	11.2	11.4	9.4	7.8	8.2	9.8	10.8	10.7	10.4
Precipitation	PrecAvg	18.9	0.7	0.5	0.8	1.3	2.4	3.2	2.8	2.3	2.3	1.4	0.7	0.6
	PrecMax	29.3	1.6	1.6	2.8	3.0	6.2	5.7	7.3	7.6	6.1	3.9	2.3	1.5
	PrecMin	11.3	0.1	0.0	0.0	0.0	0.4	0.7	0.6	0.5	0.2	0.1	0.0	0.0
	PrecStd	4.1	0.4	0.4	0.6	0.9	1.4	1.3	1.6	1.5	1.5	1.1	0.6	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	43.3	63.2	76.3	88.3	90.6	91.3	92.8	89.6	78.8	58.7	39.4
		MCWB	34.9	40.4	55.0	55.8	63.5	70.8	76.1	73.1	68.2	60.9	49.8	35.8
	2%	DB	34.4	37.0	51.9	70.2	81.3	84.4	88.1	88.1	83.5	70.1	52.2	36.3
		MCWB	33.1	35.2	46.0	52.5	59.6	67.8	74.0	71.5	67.1	58.1	46.2	34.3
	5%	DB	29.8	34.3	45.1	64.0	75.2	81.7	84.0	83.8	79.2	63.6	46.3	33.9
		MCWB	29.3	33.1	41.3	49.6	57.1	66.4	72.1	68.9	63.4	54.2	42.2	32.4
	10%	DB	26.9	29.8	39.1	59.2	71.8	78.9	81.5	81.2	73.4	58.6	42.7	30.0
		MCWB	25.6	29.0	36.7	47.5	56.0	65.1	70.4	67.6	61.8	51.9	38.9	29.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	41.0	56.8	59.0	67.6	77.8	80.8	77.8	73.5	63.8	51.8	37.1
		MCDB	37.6	44.0	60.6	69.6	76.3	83.7	86.3	86.4	82.9	71.7	57.0	38.7
	2%	WB	32.9	36.0	46.1	54.3	64.2	72.8	77.1	74.5	69.4	59.7	46.6	34.4
		MCDB	34.4	37.2	49.8	65.6	72.5	80.7	84.3	83.5	78.8	68.0	49.9	36.0
	5%	WB	28.8	33.0	41.1	51.4	61.4	69.7	74.5	72.1	66.3	55.5	42.8	31.9
		MCDB	29.2	34.3	44.9	61.5	69.9	76.9	81.4	80.3	74.6	63.0	46.9	33.4
	10%	WB	25.4	28.5	37.4	48.6	58.6	67.3	72.2	69.6	63.7	51.8	38.8	29.3
		MCDB	26.9	29.2	39.3	58.5	68.4	75.1	79.1	77.0	71.1	58.0	42.1	29.8
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.5	17.2	20.8	23.5	21.4	22.6	24.6	23.6	18.3	15.4	15.1
		MCDBR	16.9	16.3	20.1	31.2	31.3	27.1	26.4	29.1	29.9	27.2	21.4	17.4
		MCWBR	15.8	14.4	14.8	17.7	15.4	13.6	15.2	14.5	15.5	16.3	15.9	15.4
	5% WB	MCDBR	16.8	15.8	19.4	27.3	23.8	22.0	23.4	24.4	24.6	24.1	20.2	16.5
		MCWBR	15.8	14.1	14.7	16.9	14.2	13.3	15.3	14.4	14.5	15.4	15.4	14.9
Clear-Sky Solar Irradiance	taub	0.281	0.305	0.305	0.357	0.373	0.388	0.390	0.386	0.352	0.310	0.280	0.267	
	taud	2.241	2.184	2.390	2.339	2.357	2.341	2.344	2.356	2.446	2.532	2.490	2.362	
	Ebn at Noon	245	265	290	283	281	277	274	270	270	267	252	239	
	Edh at Noon	24	32	31	36	37	38	38	36	30	23	19	19	
All-Sky Solar Radiation	RadAvg	476	869	1249	1551	1740	1869	1997	1648	1205	732	419	327	
	RadStd	50	62	119	138	120	143	85	107	85	102	55	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page